Literature-Implied Partial Subcase for arXiv:0912.5113

FA/Banach run fa_banach_001

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Status

This is a `literature_implied_answer (partial subcase)` packet. It does not claim a new solution to the general Banach-space question in Baudier–Kalton–Lancien [1]. It records that a later theorem of Perreau [2] gives the desired implication for quasi-reflexive Banach spaces.

Source Problem

Baudier–Kalton–Lancien study the family $(T_N)_{N\geq 1}$ of finite countably branching trees with the hyperbolic graph metric. They prove, for separable reflexive Banach spaces X, the equivalence of:

$$\mathrm{Sz}(X) > \omega \text{ or } \mathrm{Sz}(X^*) > \omega,$$

the existence of a Lipschitz embedding of T_∞ into X, and the existence of a constant $C \geq 1$ such that $T_N \overset{C}{\hookrightarrow} X$ for every N. Immediately after this corollary, they ask whether the finite-tree condition implies the Szlenk-index condition for general Banach spaces.

Supporting Literature

Perreau's later paper extends the non-embeddability half of the Baudier–Kalton–Lancien theorem to quasi-reflexive spaces. In the notation of [2], if X is quasi-reflexive and both $\mathrm{Sz}(X) \leq \omega$ and $\mathrm{Sz}(X^*) \leq \omega$, then the family (T_N) does not equi-Lipschitz embed into X.

Definitions and Identification

Baudier–Kalton–Lancien write $M \overset{C}{\hookrightarrow} N$ when there is a Lipschitz embedding from M into N whose distortion is at most C. Perreau says that a family (M_i) equi-Lipschitz embeds into N if there are maps $f_i : M_i \to N$ and constants $a, b > 0$, independent of i, such that

$$a\, d_i(x, y) \leq \|f_i(x) - f_i(y)\| \leq b\, d_i(x, y).$$

For Banach-valued embeddings these formulations agree for a family with uniform distortion: if each T_N embeds in X with distortion at most C, rescale the Nth map so that its lower Lipschitz constant is 1; then the upper Lipschitz constant is at most C for every N.

$j=1 \quad k=1$

Now (3.14) and (3.15) give a contradiction since $m > (2C)^q$. □

As an immediate consequence of Theorem 3.1 and section 2 we obtain the following characterization, which yields Theorem 1.2 announced in our introduction.

Corollary 3.4. *Let X be a separable reflexive Banach space. The following assertions are equivalent*

(i) $Sz(X) > \omega$ or $Sz(X^*) > \omega$.

(ii) There exists $C \geq 1$ such that $T_\infty \overset{C}{\hookrightarrow} X$.

(iii) There exists $C \geq 1$ such that for any N in $\mathbb{N}$, $T_N \overset{C}{\hookrightarrow} X$.

Remark. Let us mention that we do not know if (iii) implies (i) for general Banach spaces.

4. Applications to coarse Lipschitz embeddings and uniform homeomorphisms between Banach spaces

Figure 1: Source evidence from arXiv:0912.5113, page 18: Corollary 3.4 and the following remark asking whether (iii) implies (i) for general Banach spaces.

Idea of the Proof

The source remark asks for an implication from uniform finite-tree embeddability to large Szlenk index. Perreau proves exactly the contrapositive under quasi-reflexivity: small Szlenk index for both X and X^* prevents the finite countably branching trees from embedding with uniform Lipschitz constants. Thus, in the quasi-reflexive category, the source's condition (iii) forces condition (i). The full non-quasi-reflexive case is not addressed.

Theorem 1. *Let X be a quasi-reflexive Banach space. Suppose there is a constant $C \geq 1$ such that, for every $N \in \mathbb{N}$, the countably branching tree T_N admits a Lipschitz embedding into X with distortion at most C. Then*

$$\mathrm{Sz}(X) > \omega \quad \textit{or} \quad \mathrm{Sz}(X^*) > \omega.$$

Proof. Assume, toward a contradiction, that $\mathrm{Sz}(X) \leq \omega$ and $\mathrm{Sz}(X^*) \leq \omega$. By hypothesis, for each N there is an embedding $f_N : T_N \to X$ with distortion at most C. Since X is a Banach space, we may multiply each f_N by a positive scalar. Rescaling separately for each N, we obtain maps $g_N : T_N \to X$ satisfying

$$d(s,t) \leq \|g_N(s) - g_N(t)\| \leq C\, d(s,t)$$

for all $s, t \in T_N$, with the same constants for all N. Hence the family $(T_N)_{N \geq 1}$ equi-Lipschitz embeds into X.

This contradicts Perreau's Theorem 1.1 [2], which states that for a quasi-reflexive Banach space X with $\mathrm{Sz}(X) \leq \omega$ and $\mathrm{Sz}(X^*) \leq \omega$, the family (T_N) does not equi-Lipschitz embed into X. Therefore the assumption was false, and at least one of $\mathrm{Sz}(X)$ or $\mathrm{Sz}(X^*)$ is larger than ω. □

Abstract. In this note we extend to the quasi-reflexive setting the result of F. Baudier, N. Kalton and G. Lancien concerning the non-embeddability of the family of countably branching trees into reflexive Banach spaces whose Szlenk index and Szlenk index from the dual are both equal to the first infinite ordinal ω. In particular we show that the family of countably branching trees does neither embed into the James space $\mathcal{J}_p$ nor into its dual space $\mathcal{J}_p^*$ for $p \in (1, \infty)$.

1. Introduction

Let $(T_N)_{N \geq 1}$ be the family of countably branching trees endowed with the hyperbolic distance. The main result in the present paper is the following theorem.

Theorem 1.1. *Let X be a quasi-reflexive Banach space. If the Szlenk index $S_Z(X)$ of the space X satisfies $S_Z(X) \leq \omega$ and if the Szlenk index $S_Z(X^*)$ of its dual space satisfies $S_Z(X^*) \leq \omega$, then the family $(T_N)_{N \geq 1}$ does not equi-Lipschitz embed into X.*

Let us briefly recall the context and the motivation of this theorem. In 1986, J. Bourgain gave in his paper [4] a metric invariant characterizing super-reflexivity: the non equi-Lipschitz embeddability of the family $(D_N)_{N \geq 1}$ of dyadic trees endowed with the hyperbolic distance. His result is the following.

Theorem 1.2. *Let X be a Banach space. Then X is super-reflexive if and only if the family $(D_N)_{N \geq 1}$ does not equi-Lipschitz embed into X.*

This was the first step in the so called *Ribe program* which looks for metric invariants characterizing *local properties* of Banach spaces. The reader can have a look at [16] for a detailed introduction to the Ribe program and for a survey of results in this direction. A short proof of the non-embeddability of the family of dyadic trees into a super-reflexive space was given more recently by R. Kloeckner in [9] using uniform convexity and a self-improvement argument.

In [2], F. Baudier, N. J. Kalton and G. Lancien introduced a new metric invariant in order to give a metric characterization of *asymptotic properties* of Banach spaces: the non equi-Lipschitz embeddability of the family $(T_N)_{N \geq 1}$ of countably branching trees endowed with the hyperbolic distance. The main tool in their paper is a derivation index called Slzenk index. We will introduce these objects in section 2.

Figure 2: Supporting evidence from arXiv:1906.02046, page 1: Perreau's Theorem 1.1 for quasi-reflexive Banach spaces.

Verification and Limitations

The packet uses no computational verifier. The verification is purely theorem-to-question matching:

- the source condition (iii) is a uniform distortion condition for all finite countably branching trees;
- uniform distortion into a Banach space gives Perreau's equi-Lipschitz formulation after rescaling;
- Perreau's theorem applies to quasi-reflexive Banach spaces and gives the contrapositive.

The limitation is substantial: this packet proves only the quasi-reflexive subcase of the source remark. The implication for arbitrary Banach spaces is left open here.

Novelty and Search Bounds

This is not an originality claim. Before packaging, the run's cheap indexes were searched for `0912.5113`, `1906.02046`, Perreau, countably branching trees, quasi-reflexive, Szlenk, and T_N keywords. No prior packet or attempt for this exact quasi-reflexive tree/Szlenk subcase was found. The supporting source was identified as a later literature theorem directly implying a subcase of the source remark, so the packet is placed under `solutions/literature_implied_answers/`.

Human Review Recommendation

Review as a provenance/status packet. The main points to check are the rescaling identification between the two embedding conventions and the absence of an extra hypothesis in Perreau's Theorem 1.1. Do not count this as a full answer to Baudier–Kalton–Lancien's general Banach-space question.

References

[1] F. Baudier, N. J. Kalton, and G. Lancien, *A new metric invariant for Banach spaces*, arXiv:0912.5113.

[2] Y. Perreau, *On the embeddability of the family of countably branching trees into quasi-reflexive Banach spaces*, arXiv:1906.02046.